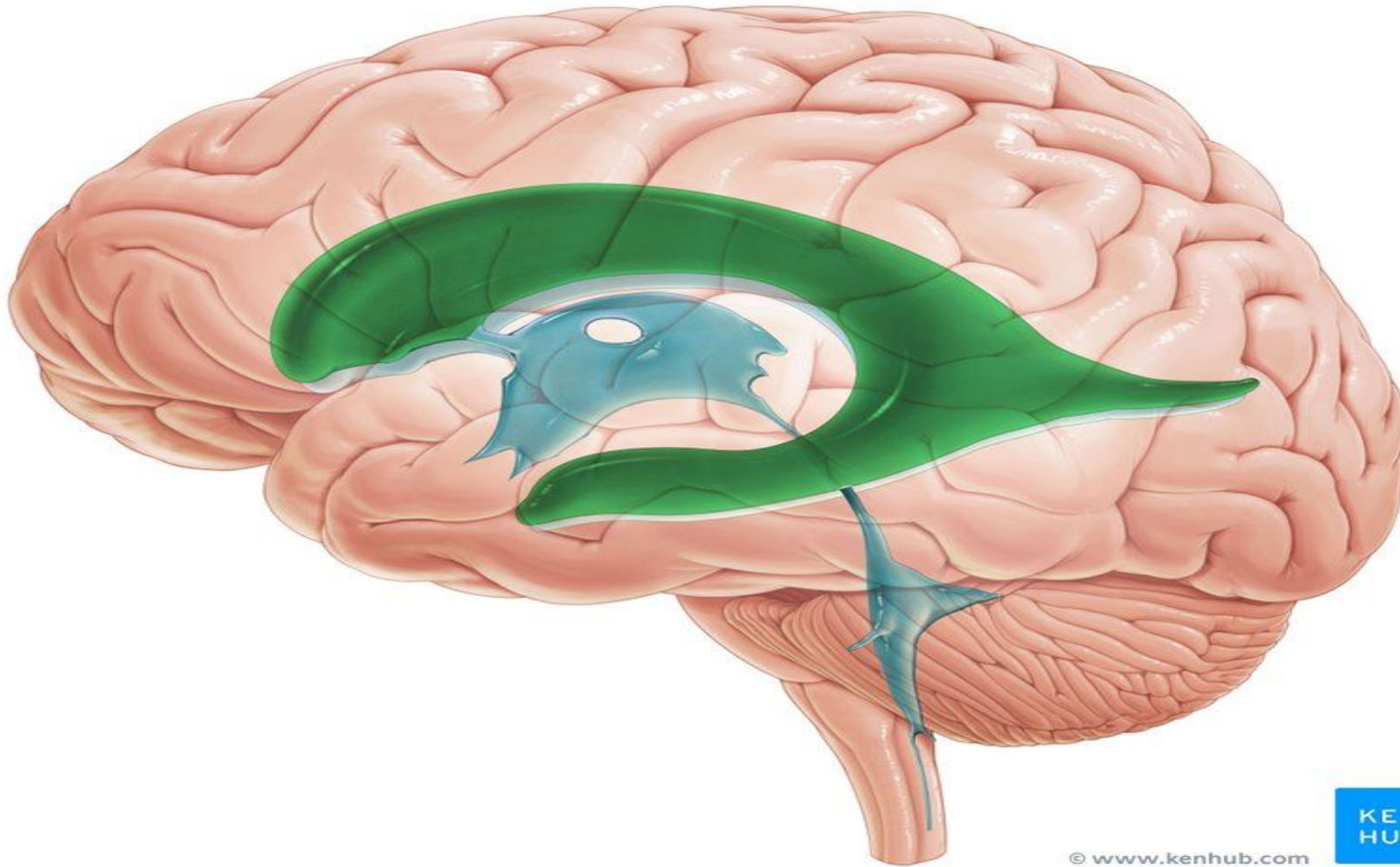


The Ventricles of Brain



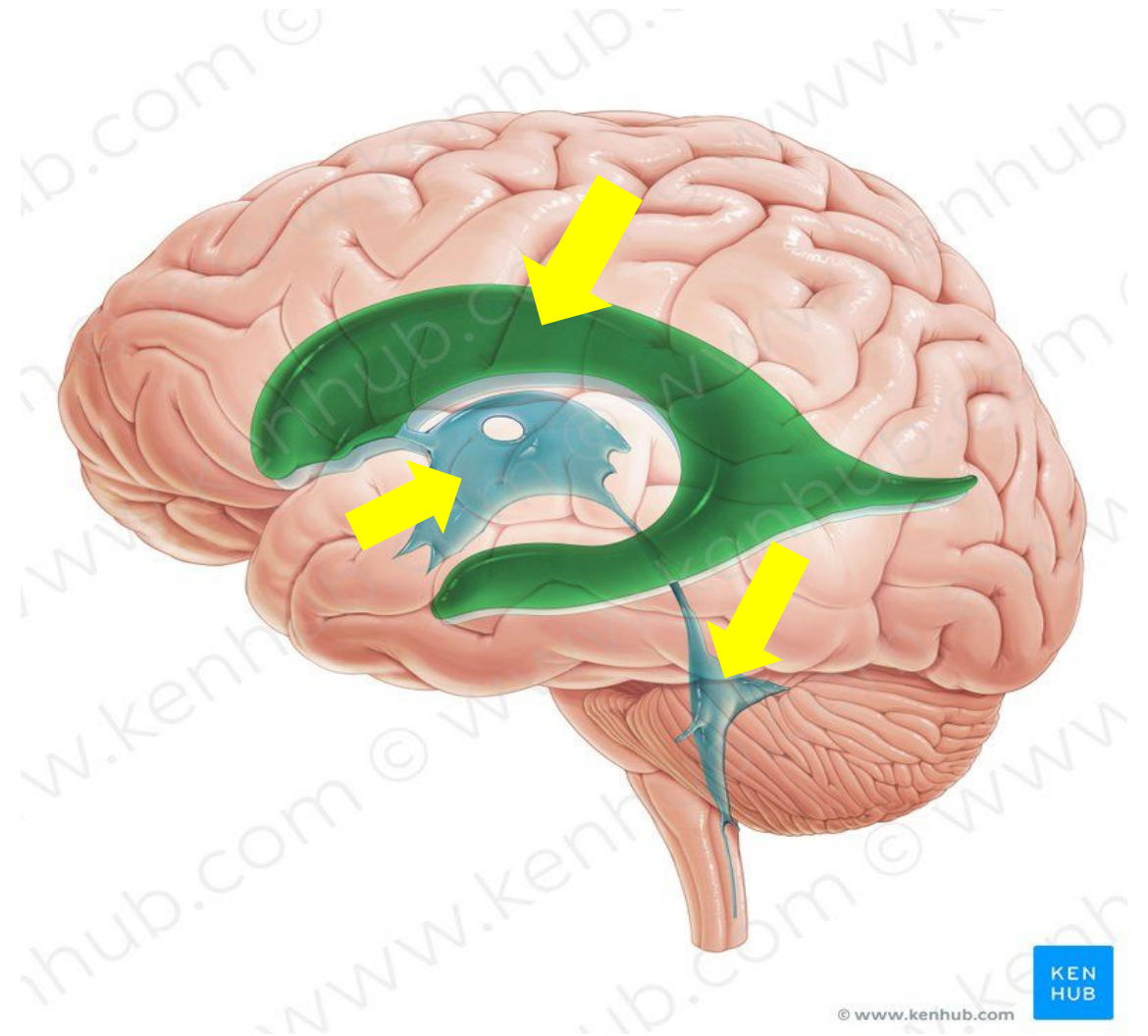
Learning Objectives:

At the end of the lecture, the students should be able to:

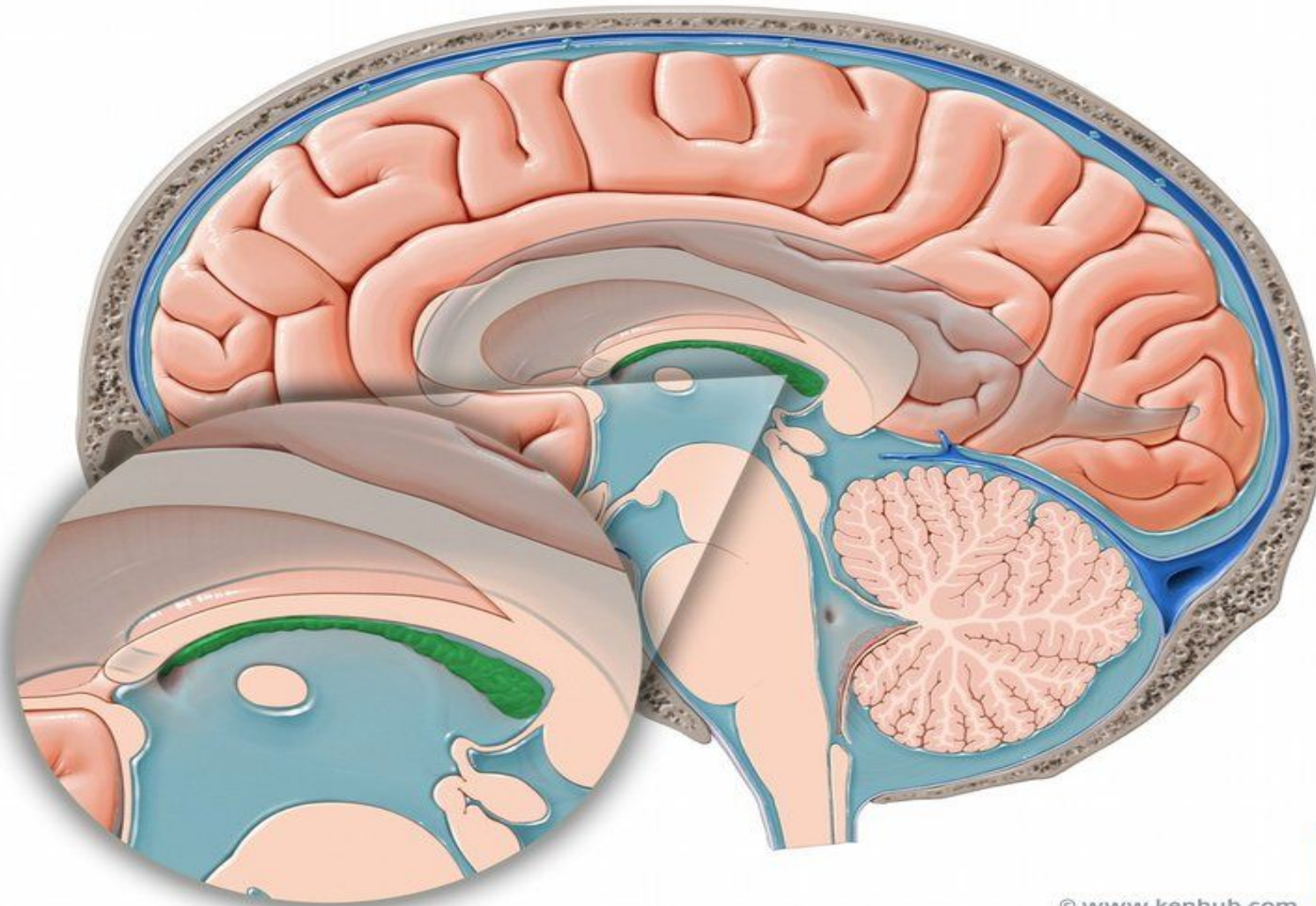
- **Define ventricles of brain.**
- **Enlist and name the ventricles of brain.**
- **Define third ventricle.**
- **Describe the anatomy (shape & boundaries) of third ventricle.**
- **Define lateral ventricles.**
- **Describe the anatomy (shape & boundaries) of different parts of lateral ventricles.**
- **Name and locate the foramina connecting the ventricles with each other.**

The Ventricular System of brain:

- The ventricles are fluid-filled cavities located within the brain.
- These are the lateral ventricles (paired), the third & the fourth ventricle(s).
- The ventricles are lined with ependyma and contain cerebrospinal fluid (CSF).
- Embryologically the ventricles are derived from the cavity of neural tube.



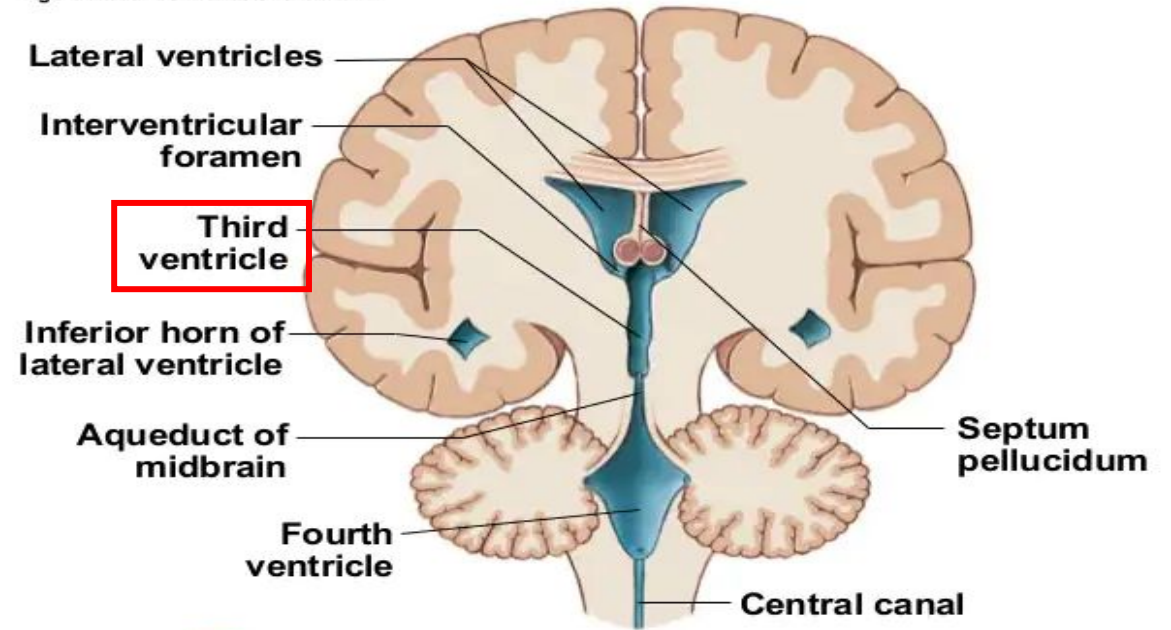
The Third Ventricle:



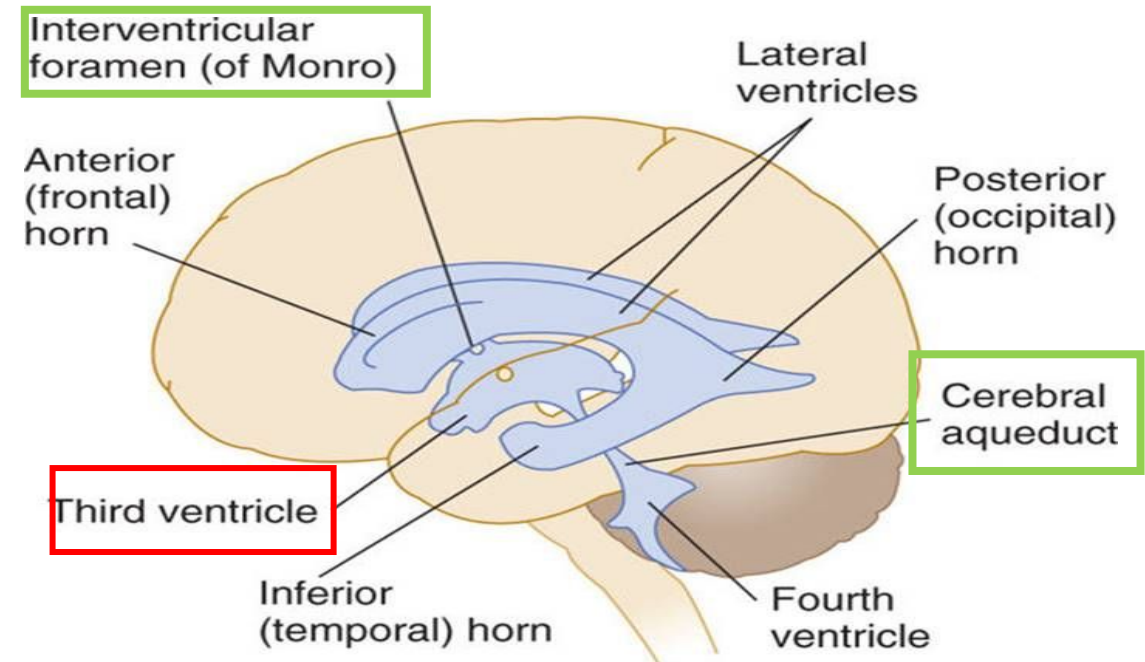
Introduction:

- Third ventricle is a slit-like cleft between two thalami.
- It is derived from forebrain vesicle.
- It communicates anteriorly with the lateral ventricles through *internventricular foramina/foramina of Monro*, &
- It communicates posteriorly with fourth ventricle through *cerebral aqueduct*.

Figure 16.2d Ventricles of the Brain

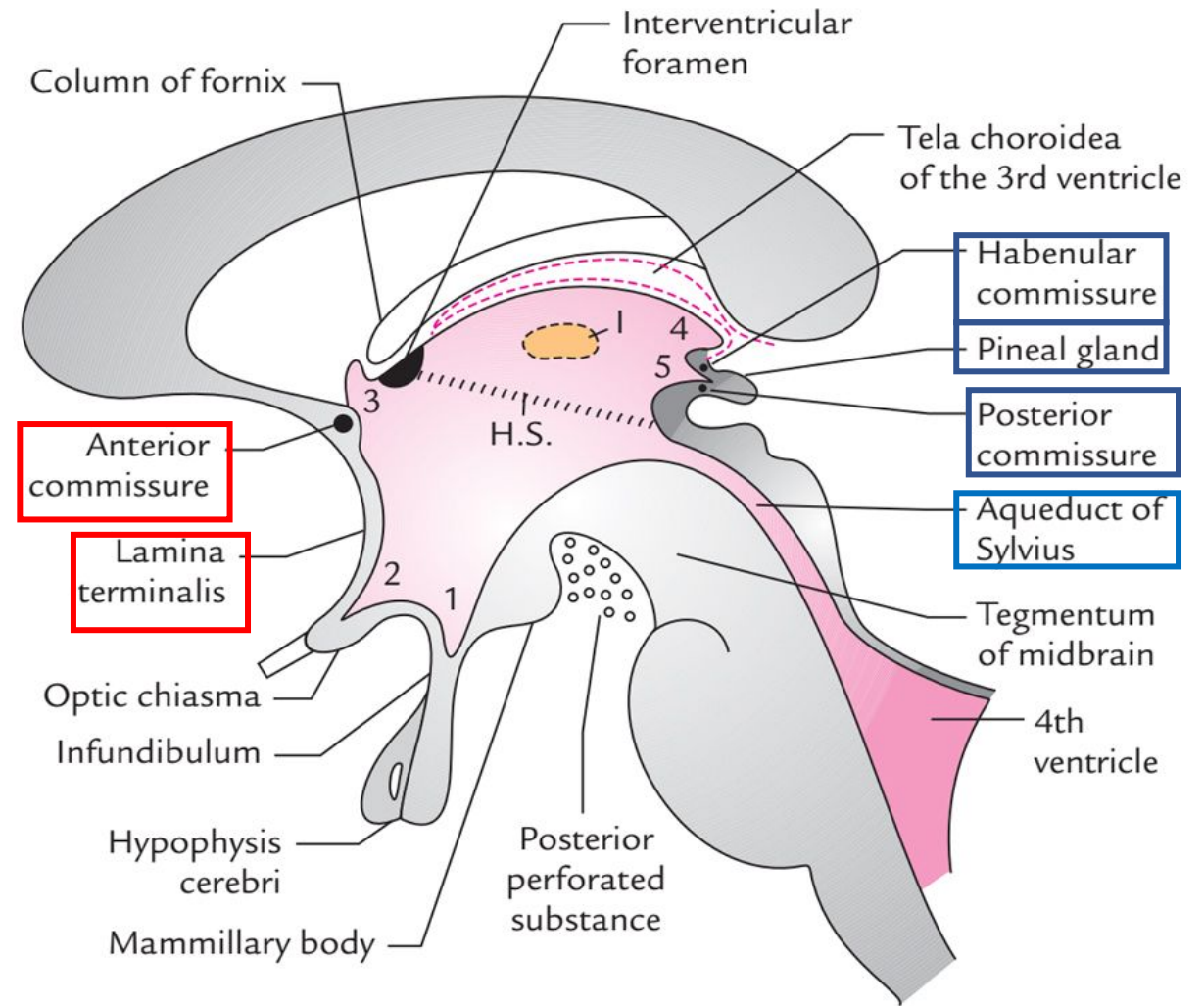
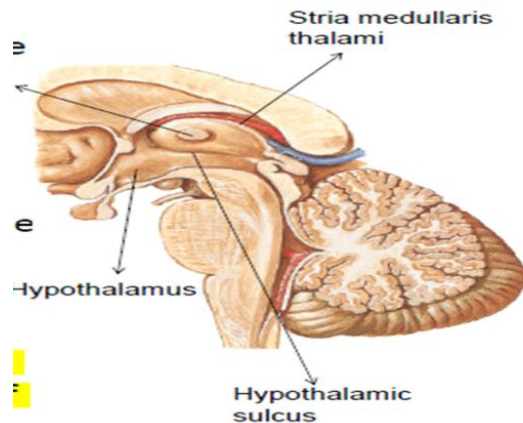


d Diagrammatic coronal section showing the interconnections between the ventricles

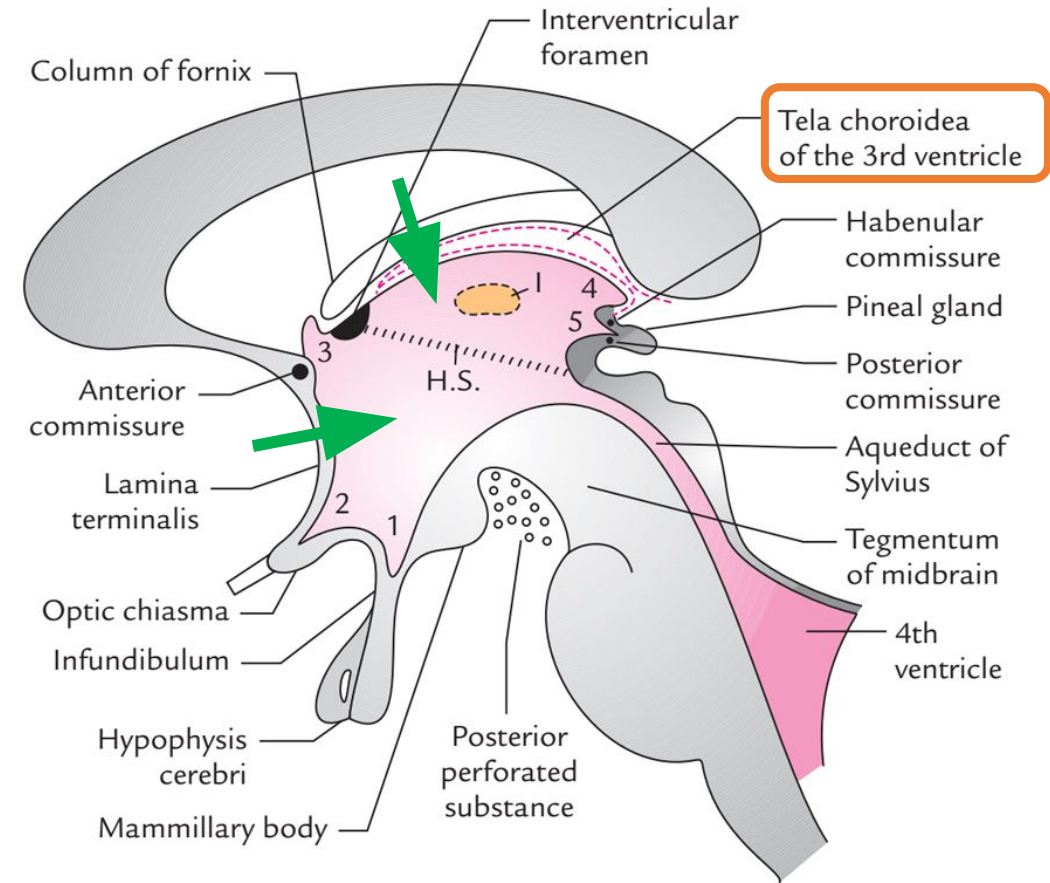


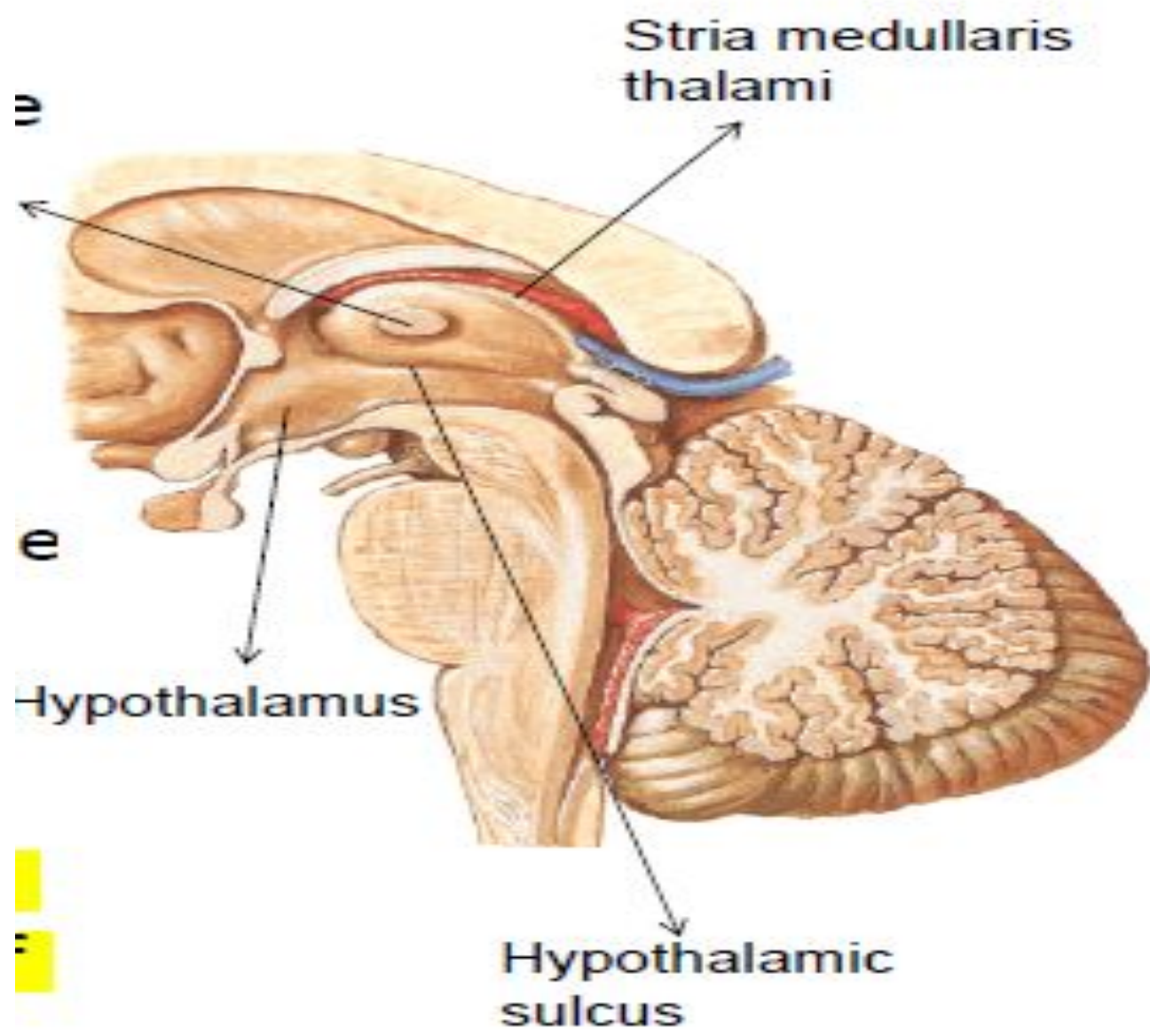
Boundaries:

- The third ventricle has anterior, posterior, lateral, superior & inferior boundaries & is lined with ependyma.
- **Anterior wall:** is formed by lamina terminalis, a thin sheet of gray matter and anterior commissure.
- **Posterior wall:** formed by opening of cerebral aqueduct. The structures superior to cerebral aqueduct in ascending order are posterior commissure, pineal gland and habenular commissure.

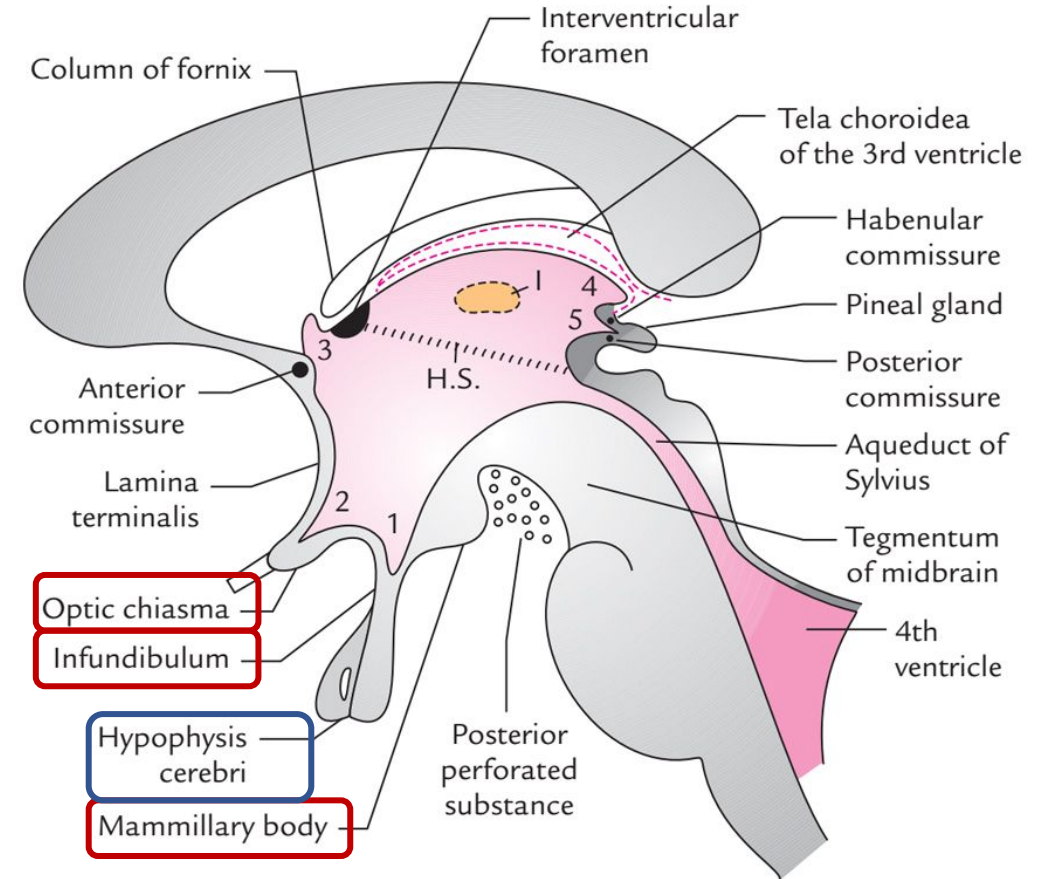


- **Lateral wall:** is formed by medial surface of thalamus superiorly & medial surface hypothalamus inferiorly, both separated by hypothalamic sulcus.
- Lateral wall is limited superiorly by stria medullaris thalami.
- **Superior wall/roof:** is formed by ependymal lining.
- Superior to this layer is tela choroidea, a two-layered fold of pia mater containing internal cerebral veins.
- Tela choroidea projects downward on each side of the midline, invaginating the ependymal roof to form choroid plexus of third ventricle.

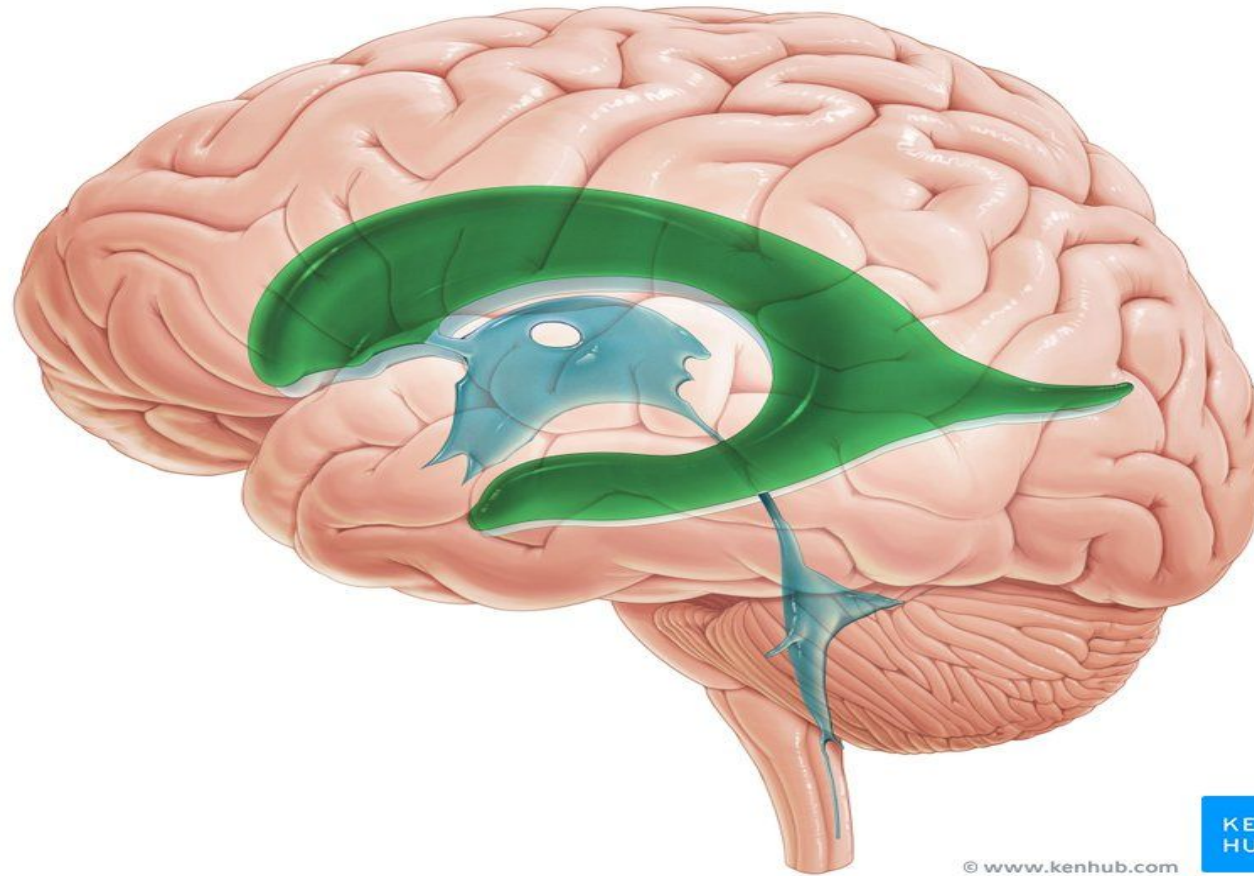


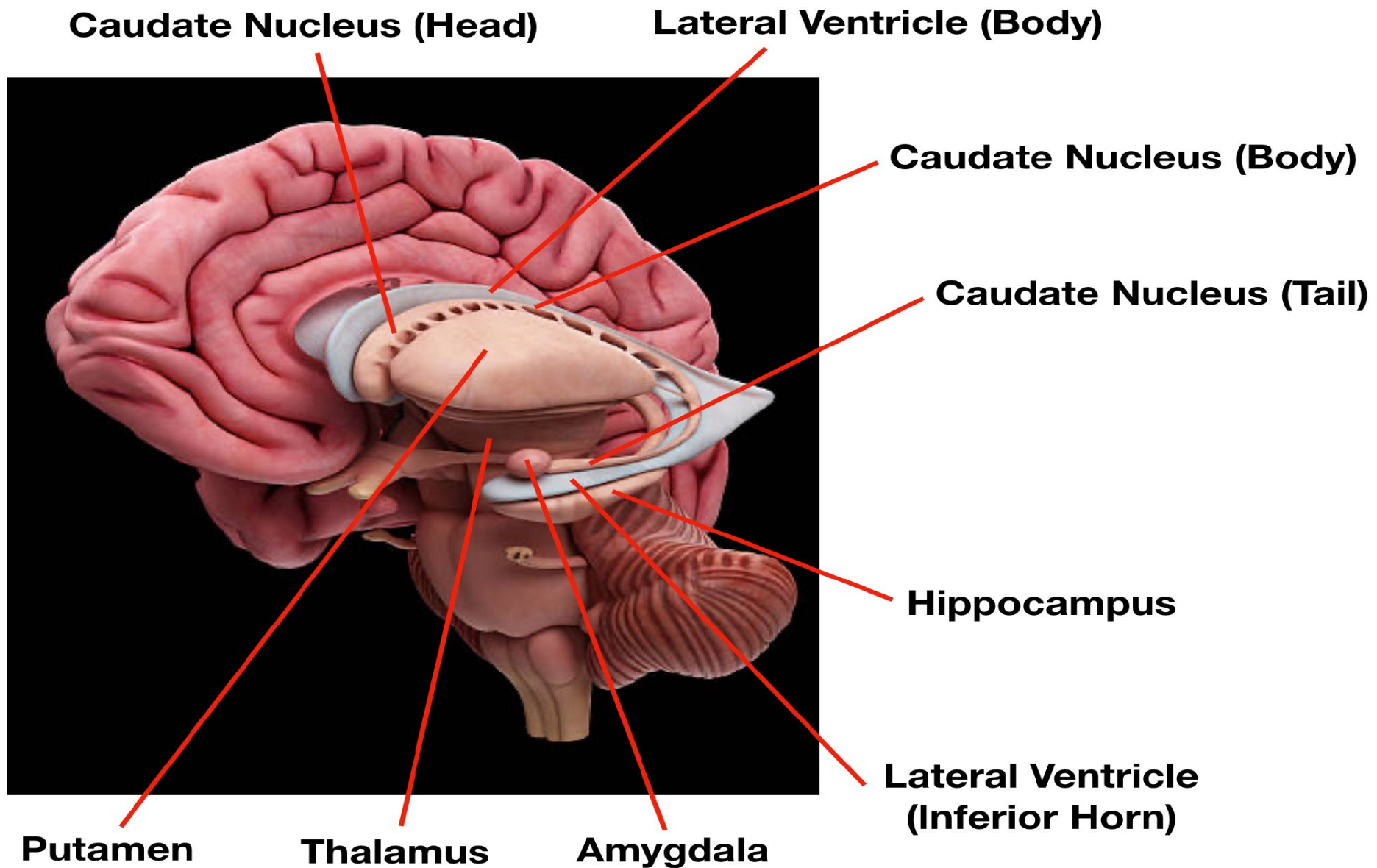


- **Inferior wall/floor:** is formed by optic chiasma, tuber cinereum, infundibulum and mamillary bodies. The hypophysis is attached to the infundibulum.



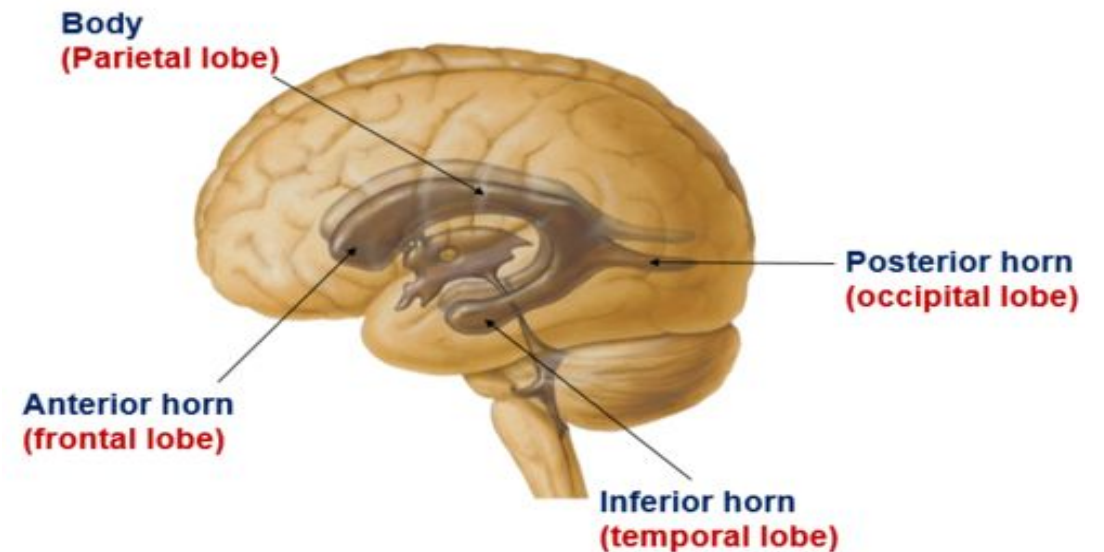
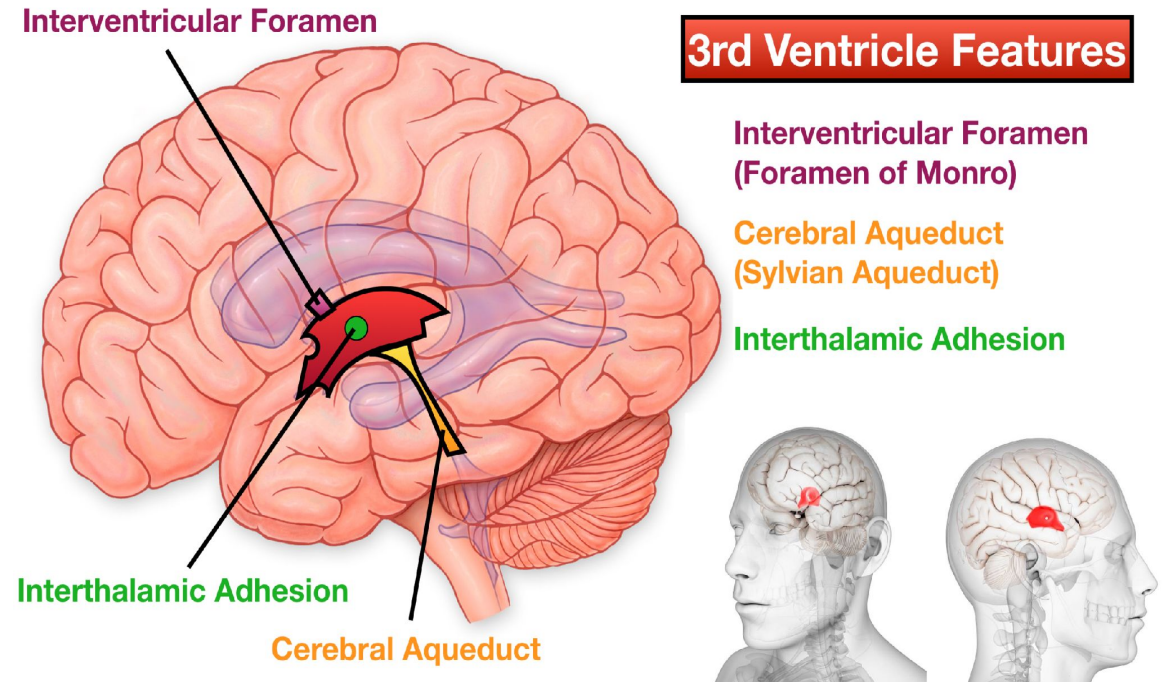
The Lateral Ventricles:





Introduction:

- There are two large lateral ventricles, one in each cerebral hemisphere.
- Each lateral ventricle is roughly C-shaped with a body, anterior, posterior and inferior horns.
- The body occupies the parietal lobe.
- From the body extend anterior, posterior & inferior horns and occupy frontal, occipital & temporal lobes respectively.
- The lateral ventricle communicates with the third ventricle through interventricular foramen.

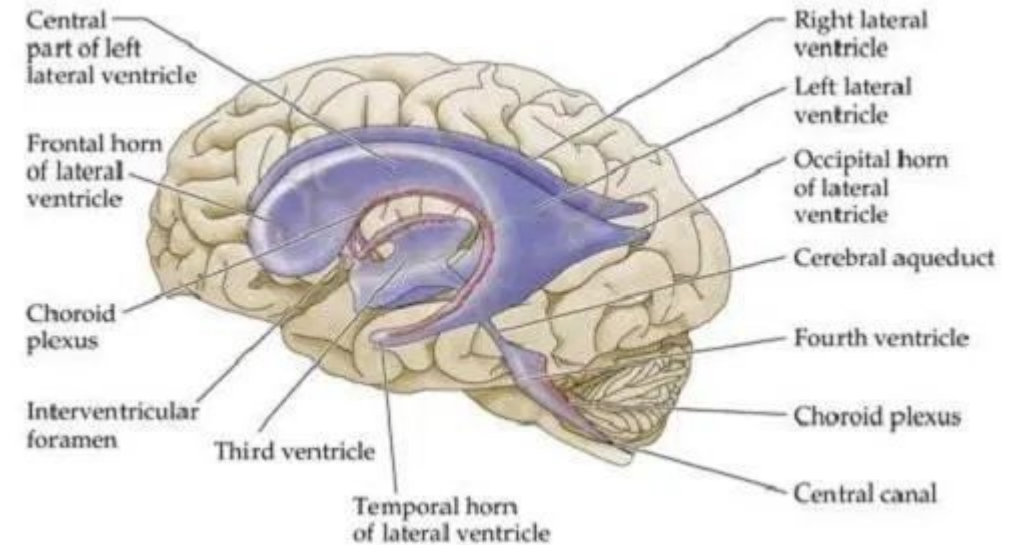


Body of lateral ventricle:

- Extends from interventricular foramen up to posterior end of thalamus where it becomes continuous with posterior and inferior horns.

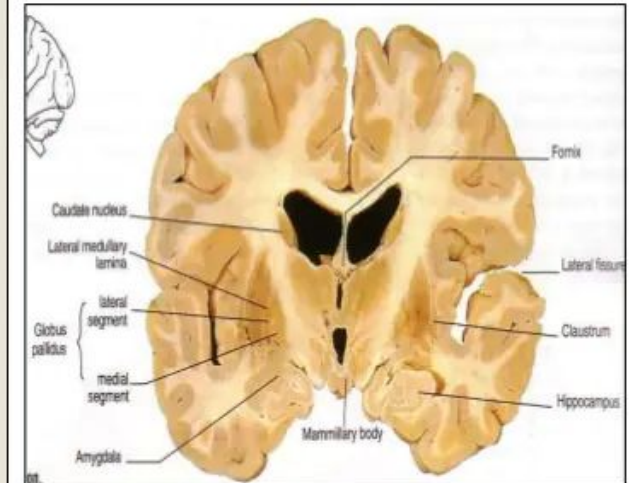
Boundaries: it has a roof, floor & medial wall.

- **Roof:** is formed by corpus callosum.
- **Floor:** is formed by body of caudate nucleus and lateral margin of thalamus.
- Choroid plexus of ventricle projects into the body through slit-like gap, the choroidal fissure between body of fornix & superior surface of thalamus.
- Through choroidal fissure blood vessels of choroid plexus invaginate into pia matter of tela choroidea & ependyma of the ventricle.
- **Medial wall:** is formed by septum pellucidum.



- Lies in the parietal lobe.
- **Roof:**
- **Corpus callosum (Trunk).**
- **Floor:**
- Sloping, From lateral to medial it is formed by:
- Body of caudate nucleus,
- Upper surface of thalamus
- Choroid plexus,
- Body of fornix.
- **Medial wall:**
- Septum pellucidum.
- **Lateral wall:**
- *narrow area at the meeting of roof & floor.*

Body or Central part

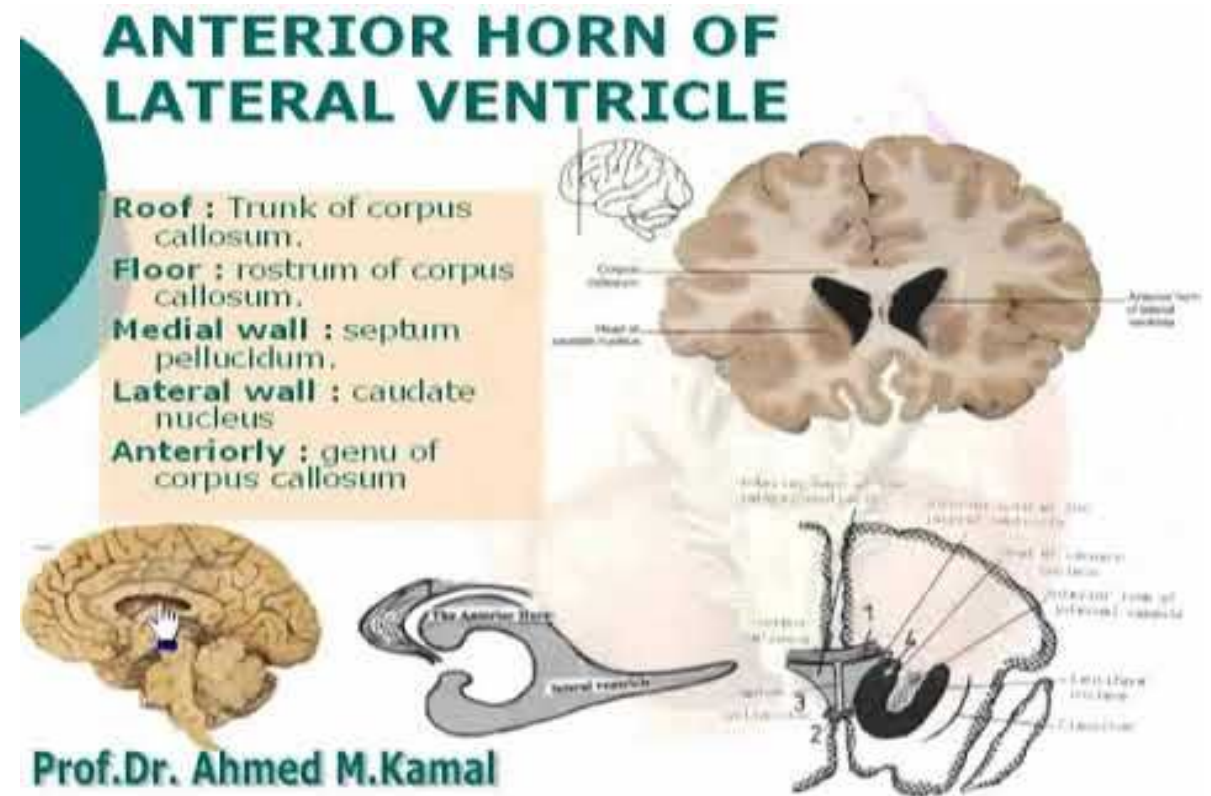
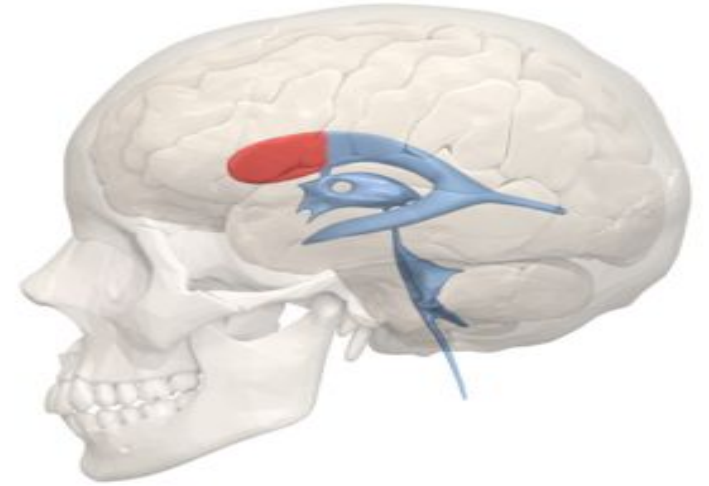


Anterior horn of lateral ventricle:

- It lies in the frontal lobe.
- It is continuous with the body at interventricular foramen.

Boundaries: it has a roof, floor & medial wall.

- **Roof:** is formed by anterior part & genu of corpus callosum.
- **Floor:** is formed by head of caudate nucleus & medially by a small portion of rostrum of corpus callosum.
- **Medial wall:** is formed by septum pellucidum & anterior column of fornix.

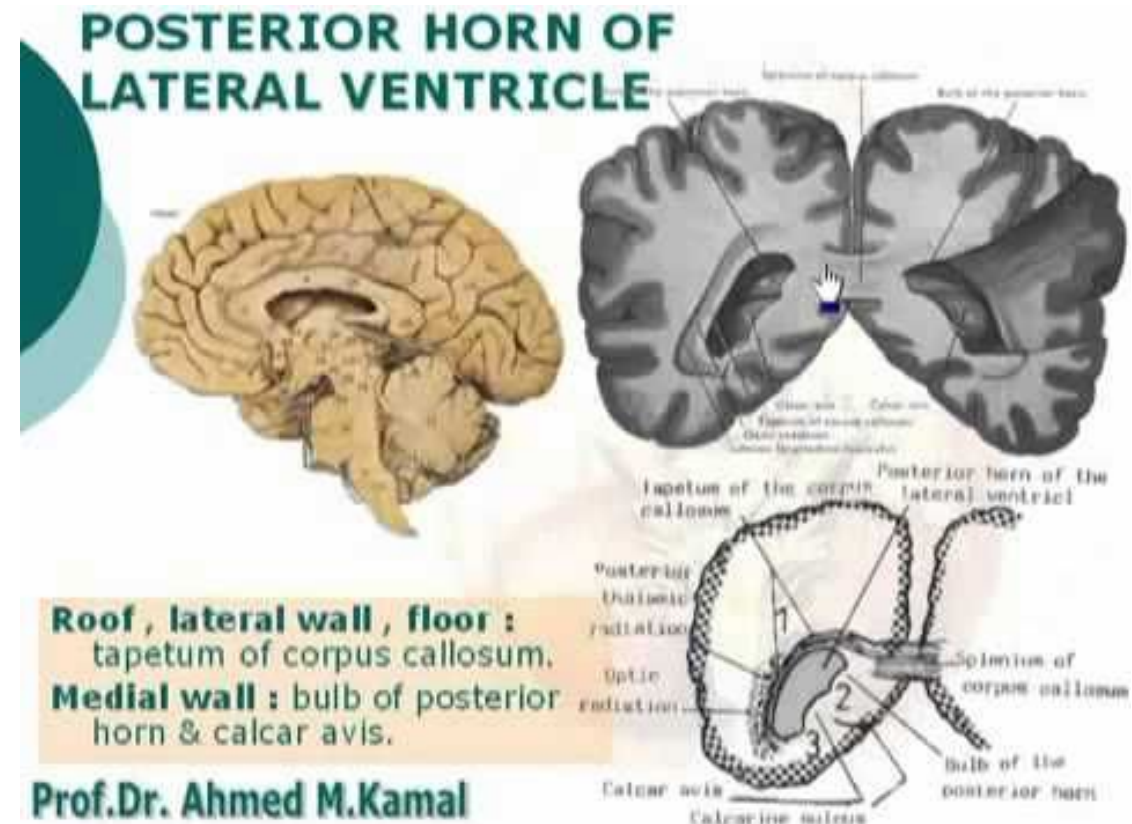
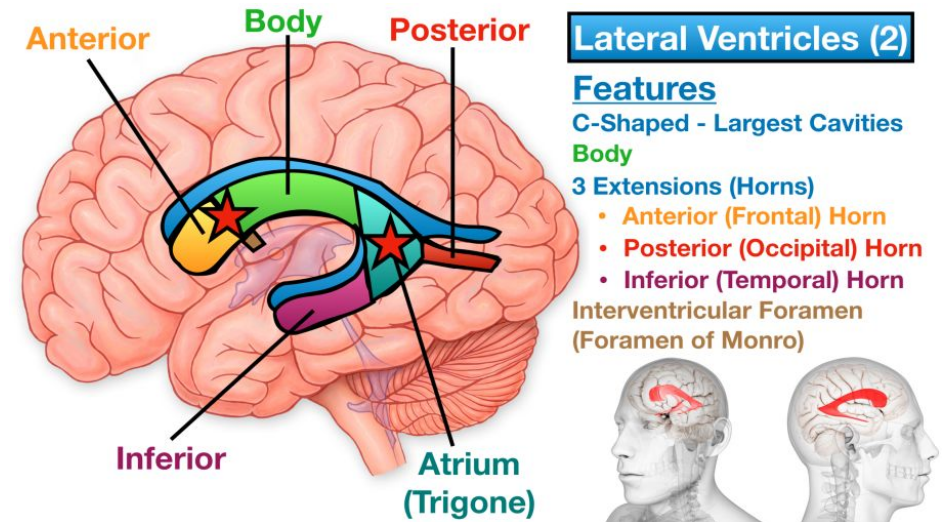


Posterior horn of lateral ventricle:

- It lies in the occipital lobe.

Boundaries: it has a roof, lateral wall & medial wall.

- **Roof & lateral wall:** are formed by fibers of tapetum of corpus callosum.
- **Medial wall:** it has two swellings, the superior swelling is called forceps major (formed by splenial fibers of corpus callosum) whereas the inferior swelling is called calcar avis produced by calcarine sulcus.

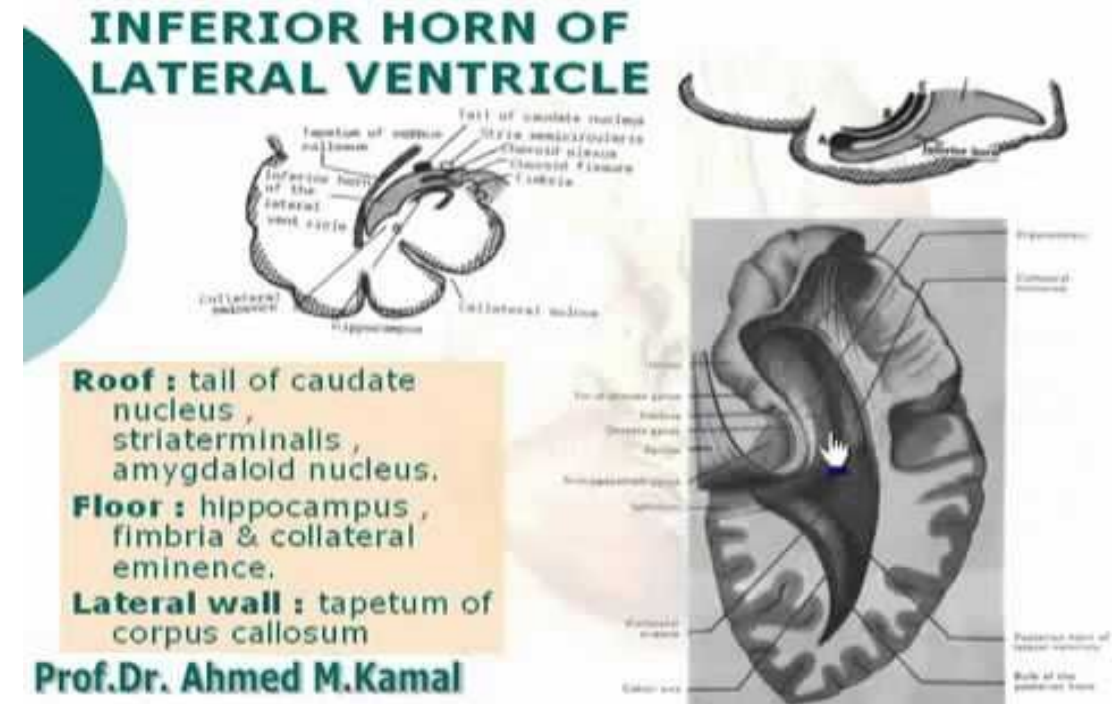
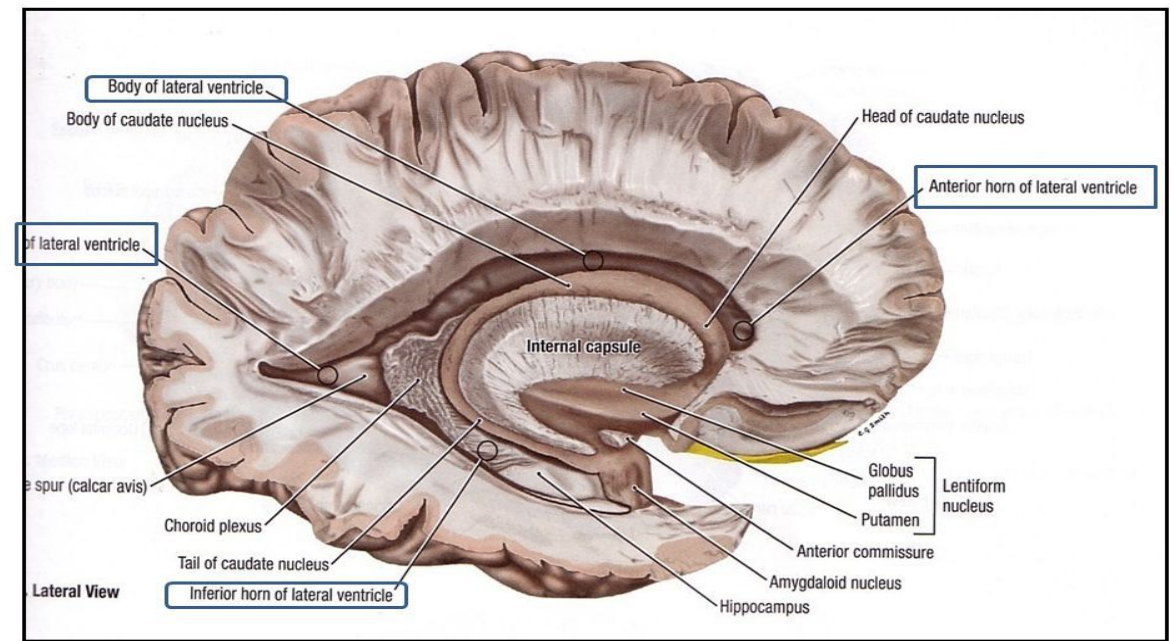


Inferior horn of lateral ventricle:

- It occupies the temporal lobe.

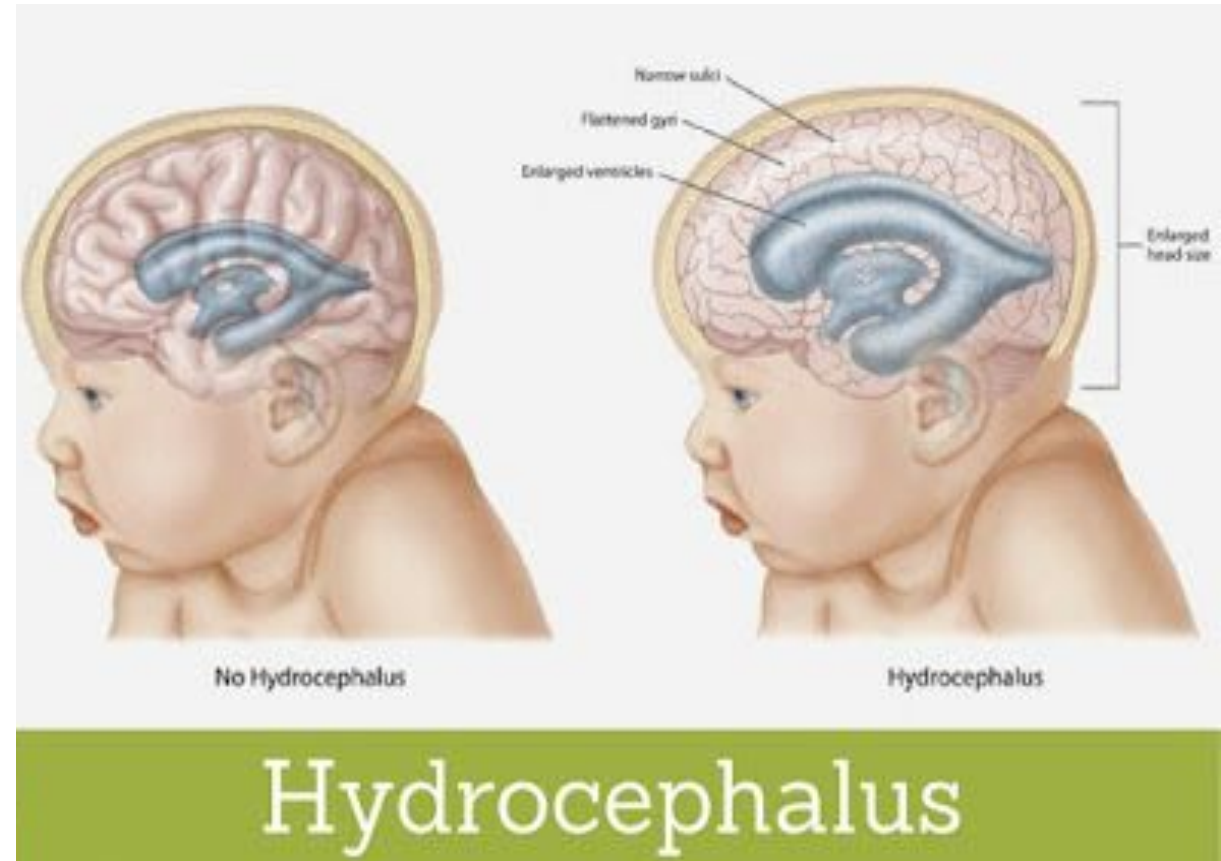
Boundaries: it has a roof & a floor.

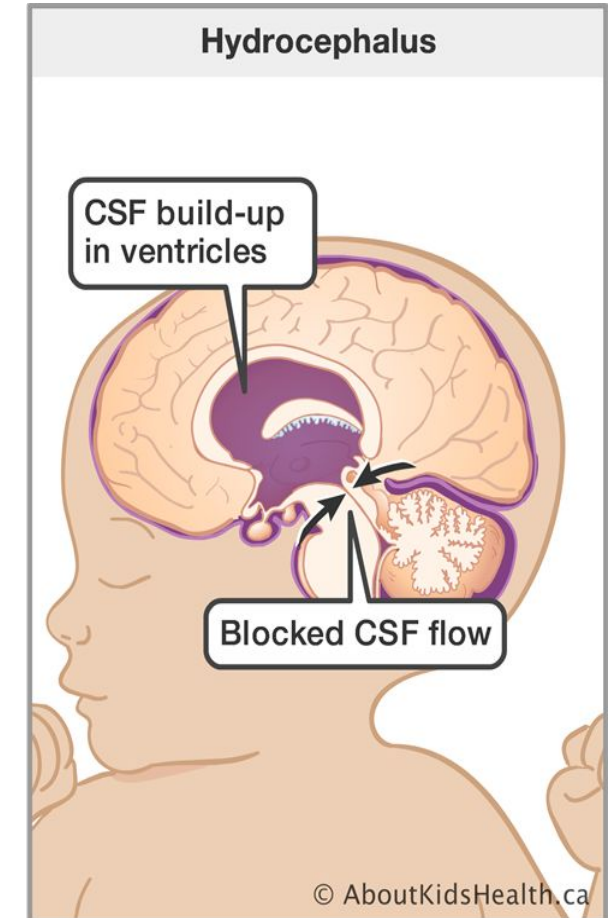
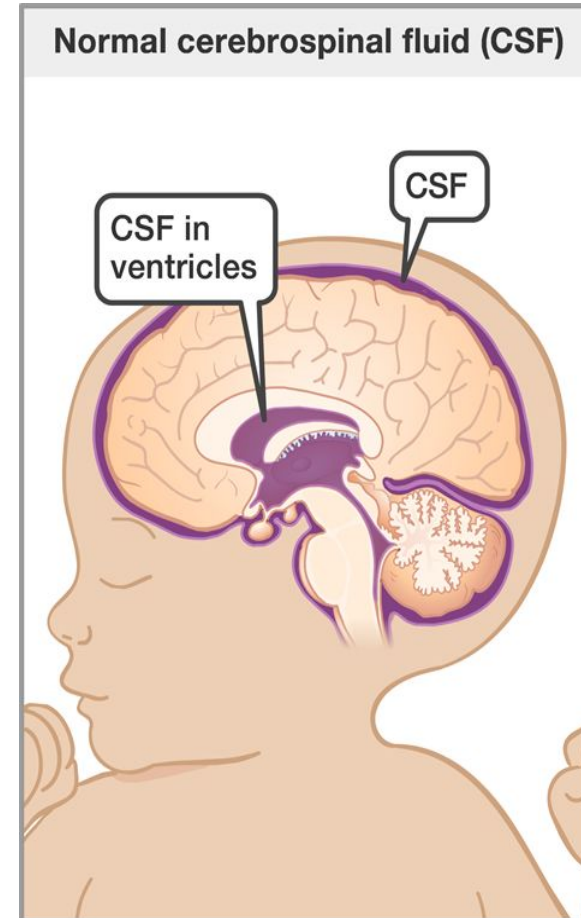
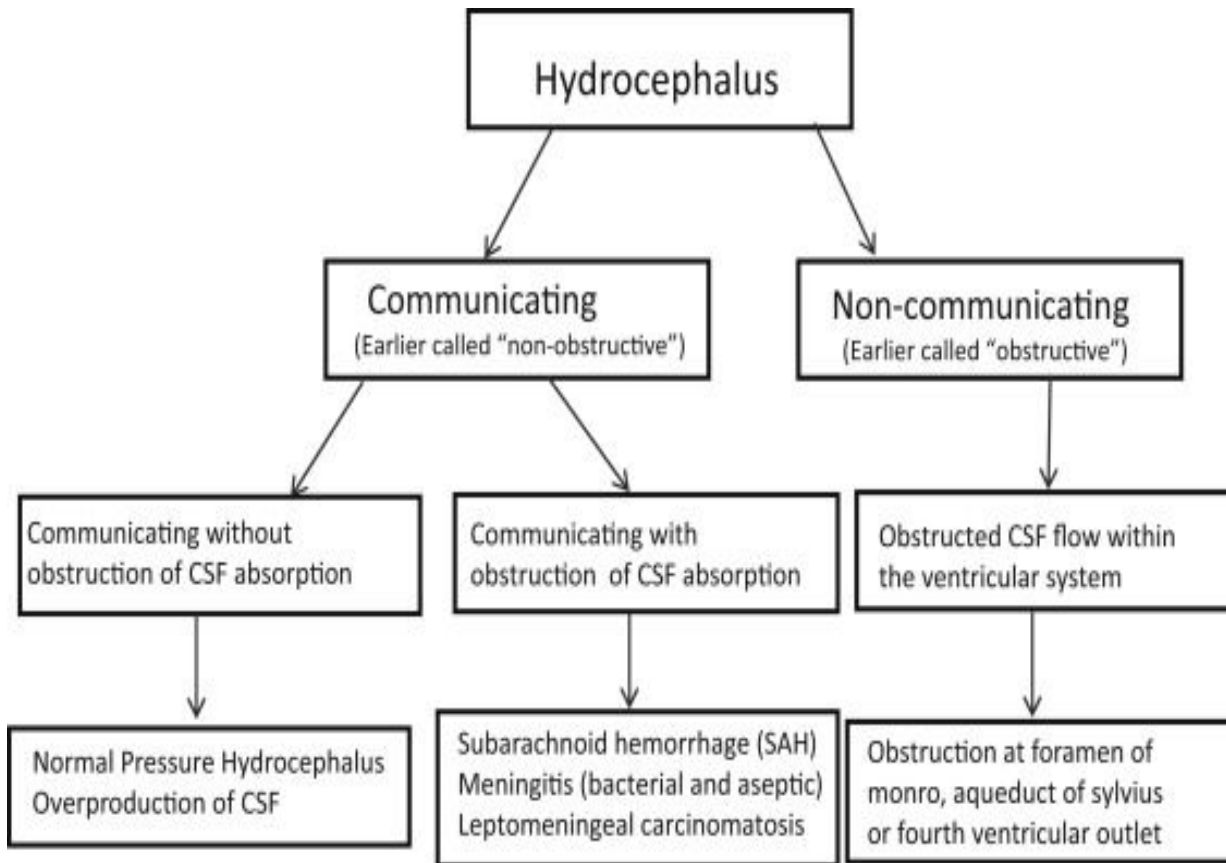
- **Roof:** is formed by tapetum of corpus callosum (inferior surface) & tail of caudate nucleus ending into amygdaloid nucleus.
- **Floor:** is formed by collateral eminence produced by collateral fissure & medially by hippocampus.



Hydrocephalus

- ✧ Abnormal dilatation of the cerebral ventricular system
- ✧ Hydrocephalus should be differentiated from disorders producing ventricular enlargement secondary to cerebral atrophy.
- ✧ Classified into :
 1. Communicating Hydrocephalus
 2. Non communicating Hydrocephalus





References:

- **CLINICAL NEUROANATOMY**

Richard S. Snell

8th edition

Chapter 16

Pages # 446-450